

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Integrates both components with at least one correct	AO1.1a	M1	$\mathbf{r} = \int 40e^{-0.2t} dt \mathbf{i} + \int 50(e^{-0.2t} - 1) dt \mathbf{j}$ $= (-200e^{-0.2t} - c) \mathbf{i} + (-250e^{-0.2t} - 50t - d) \mathbf{j}$
	Obtains correct terms. (condone missing constants)	AO1.1b	A1	$t = 0, \mathbf{r} = 0\mathbf{i} + 0\mathbf{j} \Rightarrow c = 200, d = 250$ OR
	Evaluates both constants (or uses definite integration) using 'their' expression for $\mathbf{r}$	AO3.4	M1	$\mathbf{r} = \int_0^t 40e^{-0.2t} dt \mathbf{i} + \int_0^t 50(e^{-0.2t} - 1) dt \mathbf{j}$ $= [-200e^{-0.2t} \mathbf{i} + (-250e^{-0.2t} - 50t) \mathbf{j}]_0^t$
	Obtains correct expression	AO1.1b	A1	$\mathbf{r} = 200(1 - e^{-0.2t}) \mathbf{i} + (250 - 250e^{-0.2t} - 50t) \mathbf{j}$
(b)	Forms equation to find t based on horizontal component	AO3.4	M1	$200(1 - e^{-0.2t}) = 100$ $t = 5 \ln 2 = 3.4657$ $y = 250 - 250e^{-0.2 \times 5 \ln 2} - 50 \times 5 \ln 2$ $= -48.3$
	Obtains correct time	AO1.1b	A1	
	Substitutes 'their' time into vertical component	AO1.1a	M1	
	Obtains correct displacement for 'their' time (to 1 sf, must have metres) FT only if both M1 marks have been awarded	AO3.2a	A1F	The parachutist has a vertical displacement of 50 m below the origin
(c)	Identifies vertical component of the velocity as first step	AO2.4	M1	$\frac{d}{dt} (50(e^{-0.2t} - 1)) = -10e^{-0.2t}$
	Differentiates vertical component of velocity correctly	AO1.1b	A1	As there is no initial vertical component of velocity (and hence no air resistance) the initial acceleration is only due to gravity.
	Considers implication of initial motion and reaches correct conclusion	AO2.2a	R1	Hence g is taken as 10 m s^{-2}
				Total 11 marks

Q2.

(a) $\mathbf{r} = (-17.5\mathbf{i} - 27\mathbf{j})t + \frac{1}{2}(0.5\mathbf{i} + 0.6\mathbf{j})t^2 + (500\mathbf{i} + 200\mathbf{j})$

M1: Use of $ut + \frac{1}{2}at^2$

A1: Correct with or without the initial position. That is with the final term missing or on the wrong side.

M1A1

A1: Correct with the initial position included.

A1

OR

$\mathbf{r} = (500 - 17.5t + 0.25t^2)\mathbf{i} + (200 - 27t + 0.3t^2)\mathbf{j}$

3

(b) $200 = -17.5t + 0.25t^2 + 500$

M1: Forming equation for one component based on position of the rock and their position vector.

$0.25t^2 - 17.5t + 300 = 0$

A1: Correct quadratic equation.

M1A1

$t = 40 \text{ or } 30$

A1: At least one correct solution.

A1

$-400 = -27t + 0.3t^2 + 200$

dM1: Forming equation for the other component.

dM1

$0.3t^2 - 27t + 600 = 0$

A1: Correct equation.

A1

$t = 40 \text{ or } 50$

dM1: Obtaining one or two positive solutions.

dM1

$\therefore t = 40$

A1: Selecting 40.

A1

7

OR

$$-27 \times 40 + 0.3 \times 40^2 + 200 = -1080 + 480 + 200$$

dM1: Substituting 40 into the other component.

(dM1)

A1: Correct substitution

(A1)

$$= -400$$

dM1: Checking this component of the position vector

(dM1)

$$\therefore t = 40$$

A1: Concluding that $t = 40$

Note that alternative methods based on trial and improvement can be awarded full marks.

(A1)

Alternative methods

$$0.25t^2 - 17.5t + 300 = 0$$

Marks allocated as above

(M1A1)

$$0.3t^2 - 27t + 600 = 0$$

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$$0.55t^2 - 44.5t + 900 = 0$$

$$t = 40 \text{ or } t = 40.9$$

A1: At least one correct solution

(A1)

$$0.25 \times 40^2 - 17.5 \times 40 + 300 = 0$$

dM1: Checking one or both solutions

(dM1)

$$0.3 \times 40^2 - 27 \times 40 + 600 = 0$$

$$\therefore t = 40$$

A1: concluding $t = 40$

(A1)

$$0.05t^2 - 9.5t + 300 = 0$$

$$t = 40 \text{ or } t = 150$$

A1: At least one correct solution

(A1)

$$0.25 \times 40^2 - 17.5 \times 40 + 300 = 0$$

dM1: Checking one or both solutions

(dM1)

$$0.3 \times 40^2 - 27 \times 40 + 600 = 0$$

$$\therefore t = 40$$

(A1)

(7)

(c) Av. Velocity =
$$\frac{(200\mathbf{i} - 400\mathbf{j}) - (500\mathbf{i} + 200\mathbf{j})}{40}$$

$$= \frac{-300\mathbf{i} - 600\mathbf{j}}{40}$$

M1: Use of change in position over time, with a subtraction to obtain position. Do not award if one position is taken as the origin.

M1

A1F: Correct expression.

A1F

$$= -7.5\mathbf{i} - 15\mathbf{j}$$

A1F: Correct final answer.

A1F

Follow through on their time from part (b).

$$\text{Av Vel} = \frac{-300\mathbf{i} - 600\mathbf{j}}{t}$$

3

- (d) No – The helicopter will follow a curved path and not move along a straight line between the two positions.

B1: No.

B1: Mentions path is longer than the distance between the two points.

Only award second B1 if the candidate has stated that the two quantities are not equal.

B2,1

2

[15]

Q3.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$

M1: Using constant acceleration equation to get $\mathbf{r}$.

A1: Correct expression for $\mathbf{r}$. Allow equivalent column vector answer.

M1A1

2

(b) $3t - 0.1t^2 = 0$

$t(3 - 0.1t) = 0$

$t = 0$ or $t = 30$

M1: Putting their $\mathbf{j}$ component equal to zero to form a quadratic equation.

A1: Correct equation.

M1A1

$t = 30$ seconds

A1: For 30 seconds. No need to see $t = 0$.

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

B1: Correct expression for the velocity in terms of t .

Can be implied by subsequent working in terms of t .

B1

$0.1t - 1 = -(3 - 0.2t)$

$2 = 0.1t$

M1: For $0.1t - 1 = \pm(3 - 0.2t)$. May be with their components if velocity stated incorrectly.

A1: Correct equation.

M1A1

$t = 20$

$$A1:t = 20$$

A1

$$\mathbf{v} = \mathbf{i} - \mathbf{j}$$

$$v = \sqrt{2} = 1.41 \text{ ms}^{-1}$$

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases

If the equation in t in line 2 is not seen:

then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ and $v = 1.41$

award 4 out of 6

or

then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ award 2 out of 6

6

[11]

Q4.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$

M1: Using constant acceleration equation to get $\mathbf{r}$.

A1: Correct expression for $\mathbf{r}$. Allow equivalent column vector answer.

M1A1

2

(b) $3t - 0.1t^2 = 0$

$$t(3 - 0.1t) = 0$$

$$t = 0 \quad \text{or} \quad t = 30$$

M1: Putting their $\mathbf{j}$ component equal to zero to form a quadratic equation.

A1: Correct equation.

M1A1

$$t = 30 \text{ seconds}$$

A1: For 30 seconds. No need to see $t = 0$.

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

*B1: Correct expression for the velocity in terms of t .
Can be implied by subsequent working in terms of t .*

B1

$$0.1t - 1 = -(3 - 0.2t)$$

$$2 = 0.1t$$

M1: For $0.1t - 1 = \pm(3 - 0.2t)$. May be with their components if velocity stated incorrectly.

A1: Correct equation.

M1A1

$$t = 20$$

$$A1: t = 20$$

A1

$$\mathbf{v} = \mathbf{i} - \mathbf{j}$$

$$v = \sqrt{2} = 1.41 \text{ ms}^{-1}$$

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases

If the equation in t in line 2 is not seen:

*then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ and $v = 1.41$
award 4 out of 6*

or

then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ award 2 out of 6

6

[11]

Q5.

(a) $\mathbf{v} = (4\mathbf{i} + 0.5\mathbf{j}) + (-0.4\mathbf{i} + 0.2\mathbf{j})t$

*M1: Use of constant acceleration equation to find $\mathbf{v}$
with $\mathbf{u} \neq 0\mathbf{i} + 0\mathbf{j}$*

A1: Correct $\mathbf{v}$.

(Could be done as a column vector.)

M1A1

2

(b) (i) $\mathbf{v} = (4\mathbf{i} + 0.5\mathbf{j}) + (-0.4\mathbf{i} + 0.2\mathbf{j}) \times 22.5$

M1: Substitution of 22.5 into their expression for the velocity, even if no marks awarded in part (a).

M1

$$= -5\mathbf{i} + 5\mathbf{j}$$

A1: Correct velocity CAO
(Could be done using column vectors.)

A1

2

(ii) North-west

B1: Correct statement of direction. Accept 315° .
Must follow from correct answer to (b)(i).

B1

1

(c) $(\mathbf{v} =)(4 - 0.4t)\mathbf{i} + (0.5 + 0.2t)\mathbf{j}$

B1: Grouping $\mathbf{i}$ and $\mathbf{j}$ components at some point in the solution.

(Could be done using column vectors.)

Allow $5 = (4 - 0.4t)\mathbf{i} + (0.5 + 0.2t)\mathbf{j}$

B1

$$5^2 = (4 - 0.4t)^2 + (0.5 + 0.2t)^2$$

M1: Seeing both components of their velocity squared and added

A1: Correct equation. (Condone including $\mathbf{i}$ and $\mathbf{j}$.)

For example: $5 = (4 - 0.4t)\mathbf{i}^2 + (0.5 + 0.2t)\mathbf{j}^2$ scores B1M1A0

$5^2 = (4 - 0.4t)\mathbf{i}^2 + (0.5 + 0.2t)\mathbf{j}^2$ scores B1M1A1

M1A1

$$0.2t^2 - 3t - 8.75 = 0$$

A1: Any correct simplified quadratic equation, with exactly three terms.

A1

$$t^2 - 15t - 43.75 = 0$$

$$t = 17.5 \text{ or } t = -2.5$$

dM1: Solving the quadratic equation.

(Allow one substitution error in correctly quoted formula)

Candidates with an incorrect quadratic equation must show method to get dM1.

dM1

$$t = 17.5$$

A1: Correct positive solution stated.

A1

6

Q6.

(a) $\mathbf{r} = \int \mathbf{v} \, dt = (4t + t^3)\mathbf{i} + (12t - 4t^2)\mathbf{j} + \mathbf{c}$
M1: either i or j term correct.
Condone no c

M1A1

When $t = 0$, $\mathbf{r} = 5\mathbf{i} - 7\mathbf{j}$
 $\mathbf{c} = 5\mathbf{i} - 7\mathbf{j}$
Any attempt at c

M1

$\mathbf{r} = (5 + 4t + t^3)\mathbf{i} + (-7 + 12t - 4t^2)\mathbf{j}$

A1

4

(b) $\mathbf{a} = \frac{d\mathbf{v}}{dt}$
 $\mathbf{a} = 6t\mathbf{i} - 8\mathbf{j}$
M1: either term correct

M1A1

2

(c) Using $\mathbf{F} = m\mathbf{a}$
Or: using $\mathbf{F} = m\mathbf{a}$

M1

$\mathbf{F} = 2(6t\mathbf{i} - 8\mathbf{j})$
 $\mathbf{F} = 2(6t\mathbf{i} - 8\mathbf{j})$
 $= 12t\mathbf{i} - 16\mathbf{j}$
When $t = 1$, $\mathbf{F} = 12\mathbf{i} - 16\mathbf{j}$

A1

$\therefore$ Magnitude of force is $(144t^2 + 256)^{\frac{1}{2}}$ when $t = 1$
Magnitude of force is $(144 + 256)^{\frac{1}{2}}$

M1

$$= 20 \text{ N}$$

$$= 20 \text{ N}$$

A1

4

[10]

Q7.

(a) $\mathbf{v} = (0.5\mathbf{i} + 0.375\mathbf{j}) \times 20 (=10\mathbf{i} + 7.5\mathbf{j})$

M1: Calculating velocity with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$ and $t = 20$.

A1: Correct expression for velocity.

M1A1

$$v = \sqrt{10^2 + 7.5^2} = 12.5 \text{ ms}^{-1}$$

dM1: Calculating speed.

A1: Correct speed.

dM1A1

4

(b) $\tan\theta = \frac{0.5}{0.375} \text{ or } \frac{10}{7.5} \left(\text{or } \tan\theta = \frac{0.375}{0.5} \text{ or } \frac{7.5}{10} \right)$

M1: Using trig to find angle. Can be inverted as shown in brackets.

A1F: Correct trig expression with any correct equivalent fraction.

M1A1F

$$\theta = 53^\circ$$

A1: Correct angle to the nearest degree. Accept 53° .

A1

OR

$$\cos\theta = \frac{7.5}{12.5} \text{ or } \frac{0.375}{0.625} \left(\text{or } \cos\theta = \frac{10}{12.5} \right)$$

(M1A1F)

$$\theta = 53^\circ$$

Note: For 37° award M1A0A0

But for $90 - 37 = 53^\circ$ award M1A1A1.

For 127° , award M1A1A0

(A1)

OR

$$\sin \theta = \frac{10}{12.5} \text{ or } \frac{0.5}{0.625} \left(\text{or } \sin \theta = \frac{7.5}{12.5} \right)$$

Note: 53.1° as final answer scores M1A1A0

(M1A1F)

$$\theta = 053^\circ$$

Condone finding angle from acceleration or position vector.

(A1)

3

(c) $(\mathbf{r} =) \frac{1}{2} (0.5\mathbf{i} + 0.375\mathbf{j}) t^2 (= 0.25 t^2 \mathbf{i} + 0.1875 t^2 \mathbf{j})$

M1: Finding an expression for position vector in terms of t .

A1: Correct position vector.

M1A1

$$500^2 = (0.25 t^2)^2 + (0.1875 t^2)^2$$

dM1: Using distance to form an equation for t .

A1: Correct equation.

dM1A1

$$t = 4 \sqrt{\frac{500^2}{0.25^2 + 0.1875^2}} = 40 \text{ seconds}$$

A1: Correct time.

A1

OR

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$$a = 0.625$$

M1: Finding magnitude of acceleration.

A1: Correct acceleration

(M1A1)

$$500 = \frac{1}{2} 0.625 t^2$$

dM1: Using distance to form an equation for t .

A1: Correct equation.

(dM1A1)

$$t = 40$$

A1: Correct time.

(A1)

OR

$$400 = \frac{1}{2} 0.5t^2 \text{ or } 300 = \frac{1}{2} 0.375t^2$$

M1: Working with one component.

A1: Correct distance (300 or 400)

A1: Correct equation.

(M1A1)

(A1)

$$t^2 = 1600$$

dM1: Solving for t.

(dM1)

$$t = 40$$

A1: Correct t.

Note: $500 \div 12.5 = 40$ is not acceptable and scores 0

(A1)

5

[12]

Q8.

(a) $\mathbf{a} = \frac{dv}{dt}$

$$\mathbf{a} = -8e^{-2t} \mathbf{i} + (6 - 6t)\mathbf{j}$$

M1: Differentiating with either of the two components correct. Do not need to see i or j.

A1: Correct i component.

A1: Correct j component.

M1

A1

A1

3

(b) (i) Using $\mathbf{F} = m\mathbf{a}$
 $\mathbf{F} = 5 \times \{-8e^{-2t} \mathbf{i} + (6 - 6t)\mathbf{j}\}$

$$= -40e^{-2t} \mathbf{i} + (30 - 30t)\mathbf{j}$$

M1: Multiplying their acceleration by 5, even if not a vector.

A1: Correct expression.

M1

A1

2

(ii) Magnitude of $\mathbf{F}$ is

$$\{(-40)^2 + (30)^2\}^{\frac{1}{2}}$$

M1: Finding magnitude from two non-zero terms. Must add terms and square root. Condone $\{(40)^2 + (30)^2\}^{\frac{1}{2}}$

M1

$$= 50$$

A1: Correct answer only.

In this part, condone lack of negative signs in expression for force in (b) (i).

A1

2

- (c) When F acts due west, j component is zero

$$30 - 30t = 0$$

M1: Putting j component equal to zero.

M1

$$t = 1$$

A1: Correct time.

A1

2

- (d) $\mathbf{r} = -2e^{-2t} \mathbf{i} + (3t^2 - t^3) \mathbf{j} + c$

M1: Integration with either of the two components correct. Do not need to see i or j .

A1: Correct i component.

A1: Correct j component.

Condone lack of $+ c$

M1

A1

A1

When $t = 0$, $\mathbf{r} = 6\mathbf{i} + 5\mathbf{j} \therefore c = 8\mathbf{i} + 5\mathbf{j}$

dM1: Finding c using $6\mathbf{i} + 5\mathbf{j}$ and $e^0 = 1$.

dM1

$$\therefore \mathbf{r} = (8 - 2e^{-2t}) \mathbf{i} + (5 + \frac{1}{4}t^4 - t^3) \mathbf{j}$$

A1: Correct position vector.

A1

5

[14]

Q9.

- (a) Resultant = $(6\mathbf{i} - 3\mathbf{j}) + (3\mathbf{i} + 15\mathbf{j})$
 $= 9\mathbf{i} + 12\mathbf{j}$

M1: Summing the two vectors

M1

A1: Correct resultant

A1

2

(b) Magnitude = $\sqrt{9^2 + 12^2}$
= 15 N

M1: Finding magnitude with an addition sign.

M1

A1F: Correct magnitude based on their answer to part (a).

A1F

2

(c) $1.5m = 9$ $2m = 12$
or
 $m = 6 \text{ kg}$ $m = 6 \text{ kg}$

M1: Applying Newton's second law to one or both of the components.

M1

A1F: Correct mass, follow through their answer to part (a). Do not award this mark if vector division with 2 components has been used, eg

$$\frac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}} = 6 \quad \text{or } 6\mathbf{i} + 6\mathbf{j} \text{ etc without a correct previous statement gives M0A0}$$

A1F

2

(d) (i) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j})t^2$

M1: Using a constant acceleration equation to find the position vector with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$

M1

A1: Correct position vector.

A1

2

(ii) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j}) \times 2^2 = 3\mathbf{i} + 4\mathbf{j}$
 $d = \sqrt{(3)^2 + (4)^2}$

M1: Finding the position vector when
 $t = 2$
 $(\mathbf{r} = (1.5\mathbf{i} + 2\mathbf{j}) \times 2 = 3\mathbf{i} + 4\mathbf{j})$ scores M0
unless it is clear how the 2 was obtained,
possibly by a correct formula in (d) (i))

M1

$$= \sqrt{25} = 5$$

A1: Correct distance

A1

2

[10]

Q10.

(a) $\mathbf{v} = \frac{d\mathbf{r}}{dt}$

M1

$$\mathbf{v} = (e^{\frac{1}{2}t} - 8)\mathbf{i} + (2t - 6)\mathbf{j}$$

i terms

A1

j terms

A1

3

(b) (i) When $t = 3$, $\mathbf{v} = -3.52\mathbf{i}$ Papers Practice. All rights reserved.

Accept $(e^{\frac{3}{2}} - 8)\mathbf{i}$

B1

Speed is 3.52 m s^{-1}

3.5 does not give 2nd B mark

B1

2

(ii) West

B1

1

(c) $\mathbf{a} = \frac{1}{2}e^{\frac{1}{2}t}\mathbf{i} + 2\mathbf{j}$

M1A1

When $t = 3$, $\mathbf{a} = \frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j}$ or $2.24\mathbf{i} + 2\mathbf{j}$

A1

3

(d) Using $\mathbf{F} = m\mathbf{a}$:

Accept $\mathbf{F} = 7\mathbf{a}$

M1

$$\mathbf{F} = 7 \left(\frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j} \right)$$

$\therefore$ Magnitude of force is $7 \left(\left(\frac{1}{2}e^{\frac{3}{2}} \right)^2 + 2^2 \right)^{\frac{1}{2}}$

M1

$\mathbf{F} = 21.025$

$\mathbf{F} = 21.0$

Accept 21

A1

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[12]

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Q11.

(a) $\mathbf{v} = (-2\mathbf{i} + 2\mathbf{j}) + (0.25\mathbf{i} + 0.3\mathbf{j}) \times 20$

M1: Finding velocity using $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

A1: Correct expression

A1

$\mathbf{v} = 3\mathbf{i} + 8\mathbf{j}$

A1: Correct velocity in simplest form

A1

3

(b) $-2 + 0.25t = 0$

M1: One component equal to zero (either i or j component).

A1: Correct equation

M1A1

$t = 8 \text{ s}$

A1: Correct time

A1

$\mathbf{v} = (2 + 0.3 \times 8)\mathbf{j} = 4.4\mathbf{j}$

A1: Correct velocity

A1

4

(c) $\mathbf{r} = (-2\mathbf{i} + 2\mathbf{j}) \times 20 + \frac{1}{2} (0.25\mathbf{i} + 0.3\mathbf{j}) \times 20^2 + (9\mathbf{i} + 7\mathbf{j})$

M1: Finding position vector using a constant acceleration equation with or without the initial position with $t = 20$.

A1: Correct expression for position vector including initial position.

M1A1

OR

$$\mathbf{r} = \frac{1}{2} ((-2\mathbf{i} + 2\mathbf{j}) + (3\mathbf{i} + 8\mathbf{j})) \times 20 + (9\mathbf{i} + 7\mathbf{j})$$

$$\mathbf{r} = 19\mathbf{i} + 107\mathbf{j}$$

A1: Correct position vector in simplest form.

A1

3

(d) $\mathbf{v}_{\text{AVERAGE}} = \frac{(19\mathbf{i} + 107\mathbf{j}) - (9\mathbf{i} + 7\mathbf{j})}{20}$

M1: Finding average velocity based on change of position. Subtraction of initial position must be seen or implied. Division by 8 scores M0

M1

$$= \frac{10\mathbf{i} + 100\mathbf{j}}{20}$$

$$= 0.5\mathbf{i} + 5\mathbf{j}$$

A1F: Correct average velocity. Follow through incorrect answers from part (c).

Allow $\frac{\mathbf{u} + \mathbf{v}}{2}$

A1F

2

[12]

Q12.

(a) $4\mathbf{i} = 5\mathbf{j} + 40\mathbf{a}$ **AG**

$$\mathbf{a} = \frac{4\mathbf{i} - 5\mathbf{j}}{40} = 0.1\mathbf{i} - 0.125\mathbf{j}$$

Forming a vector equation based on constant acceleration

M1

Correct equation

A1

Solving for a

dM1

Correct a from correct working

A1

For $\frac{4\mathbf{i} - 5\mathbf{j}}{40}$ on its own give M0
Allow verification

4

(b) $\mathbf{r} = 5\mathbf{j} \times 40 + \frac{1}{2} (0.1\mathbf{i} - 0.125\mathbf{j}) \times 40^2$
 $= 80\mathbf{i} + 100\mathbf{j}$

Finding position vector

M1

Correct expression

A1

Correct simplified result

A1

3

(c) (i) $\mathbf{v} = 5\mathbf{j} + (0.1\mathbf{i} - 0.125\mathbf{j})t$
 $= 0.1t\mathbf{i} + (5 - 0.125t)\mathbf{j}$

Expression for v

M1

Correct expression for v seen or implied

A1

$$5 - 0.125t = -0.1t$$

$$5 = 0.025t$$

Equating components, with or without a minus sign

dM1

Correct equation

A1

$$t = \frac{5}{0.025} = 200$$

Correct time.

A1

5

$$(ii) \quad \mathbf{v} = 0.1 \times 200\mathbf{i} + (5 - 0.125 \times 200)\mathbf{j} \\ = 20\mathbf{i} - 20\mathbf{j}$$

Finding velocity using their time

M1

Correct velocity for their time

A1F

2

Note: Consistent use of $\mathbf{u} = 4\mathbf{i}$ or $5\mathbf{i}$ or $\mathbf{a} = 0.1\mathbf{i} + 0.125\mathbf{j}$ award method marks only.

[14]

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Q13.

$$(a) \quad \mathbf{v} = \frac{d\mathbf{r}}{dt}$$

$$\mathbf{v} = (3t^2 - 6t)\mathbf{i} + (4 + 2t)\mathbf{j}$$

M1A1

2

$$(b) \quad (i) \quad \mathbf{a} = (6t - 6)\mathbf{i} + 2\mathbf{j}$$

M1A1ft

Using $\mathbf{F} = m\mathbf{a}$:

$$\mathbf{F} = (18t - 18)\mathbf{i} + 6\mathbf{j}$$

A1ft

3

(ii) When $t = 3$, $\mathbf{F} = 36\mathbf{i} + 6\mathbf{j}$

Magnitude is $\sqrt{36^2 + 6^2}$

M1

$= 36.5$

Accept $6\sqrt{37}$; ft from (b)(i)

A1ft

2

(c) When $\mathbf{F}$ acts due north:
Component of $\mathbf{F}$ in the $\mathbf{i}$ direction is 0

M1

$$18t - 18 = 0$$

$$t = 1$$

ft from (b)(i)

A1ft

2

[9]

Q14.

Use of column vectors is acceptable throughout this question.

(a) $\mathbf{v} = 20\mathbf{i} + (-0.4\mathbf{i} + 0.5\mathbf{j})t$

Use of constant acceleration equation to find expression for $\mathbf{v}$

M1

Any correct expression.

A1

2

(b) $\mathbf{v} = (20 - 0.4t)\mathbf{i} + 0.5t\mathbf{j}$

*Simplifying $\mathbf{v}$. (May be implied.)
(Missing brackets may be condoned if followed by correct working.)*

M1

$$20 - 0.4t = 0$$

Putting $\mathbf{i}$ component equal to zero

m1

$$t = \frac{20}{0.4} = 50 \text{ seconds}$$

Correct time

Candidates who are able to see the correct time without supporting working gain full marks.

Condone $\frac{20\mathbf{i}}{0.4\mathbf{i}} = 50$

A1

3

(c) $\mathbf{r} = 20\mathbf{i} \times t + \frac{1}{2} (-0.4\mathbf{i} + 0.5\mathbf{j}) \times t^2$

Use of constant acceleration equation to find expression for $\mathbf{r}$

M1

Any correct expression

A1

2

(d) (i) $\mathbf{r} = 20\mathbf{i} \times 100 + \frac{1}{2} (-0.4\mathbf{i} + 0.5\mathbf{j}) \times 100^2$

Substituting $t = 100$ into their expression for $\mathbf{r}$ (dependent on M1 in part (c))

m1

$$= 2000\mathbf{i} - 2000\mathbf{i} + 2500\mathbf{j}$$

$$= 2500\mathbf{j}$$

Correct simplified position vector ie $2500\mathbf{j}$

A1

Therefore due north

Conclusion that helicopter is due north provided their position vector is of the form $k\mathbf{j}$, where $k > 0$

Note if integration is used there is no need to prove that the constant is zero.

Note marks for (d) (i) can be awarded if part (c) scores zero.

A1

3

(ii) $\mathbf{v} = (20 - 0.4 \times 100)\mathbf{i} + 0.5 \times 100\mathbf{j}$

Substituting $t = 100$ into their expression for $\mathbf{v}$ (dependent on M1 in part (a)) or use of other constant acceleration equation and their position vector (dependent on M1 in part (c))

m1

$$= -20\mathbf{i} + 50\mathbf{j}$$

Correct simplified velocity

A1

$$v = \sqrt{20^2 + 50^2} = 53.9$$

Correct speed (Accept $10\sqrt{29}$)

A1

Note marks for (d) (ii) can be awarded if part (a) scores zero.

3

[13]

Q15.

$$(a) \quad 75\mathbf{i} = (5\mathbf{i} - 2\mathbf{j}) \times 10 + \frac{1}{2} \mathbf{a} \times 10^2$$

Equation to find $\mathbf{a}$ from $\mathbf{r} = \mathbf{ut} + \frac{1}{2} \mathbf{at}^2$

M1

Correct expression

A1

$$\mathbf{a} = \frac{75\mathbf{i} - 50\mathbf{i} + 20\mathbf{j}}{50} = 0.5\mathbf{i} + 0.4\mathbf{j}$$

AG Correct $\mathbf{a}$ from correct working

A1

3

$$(b) \quad \mathbf{r} = (5\mathbf{i} - 2\mathbf{j}) \times 8 + \frac{1}{2} (0.5\mathbf{i} + 0.4\mathbf{j}) \times 8^2$$

Expression for $\mathbf{r}$ using $t = 8$ with no extra terms

M1

Correct expressions

A1

$$= 56\mathbf{i} - 3.2\mathbf{j}$$

Correct position vector

A1

3

(c) $\mathbf{v} = (5 + 0.5t)\mathbf{i} + (0.4t - 2)\mathbf{j}$

Expression for $\mathbf{v}$. Correct expression

M1A1

$$0.4t - 2 = 0$$

$\mathbf{j}$ component equal to zero

dM1

$$t = \frac{2}{0.4} = 5$$

Correct t

A1

$$\mathbf{r} = (5\mathbf{i} - 2\mathbf{j}) \times 5 + \frac{1}{2} (0.5\mathbf{i} + 0.4\mathbf{j}) \times 5^2$$

Expression for $\mathbf{r}$ using t from $\mathbf{j}$ component equal to zero

dM1

$$= 31.25\mathbf{i} - 5\mathbf{j}$$

$$= 31.3\mathbf{i} - 5\mathbf{j}$$

Correct position vector

A1

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6

[12]

Q16.

(a) $\mathbf{v} = 1.2\mathbf{i} + (-0.9 \sin t)\mathbf{j}$

Convincingly differentiation

M1A1

2

(b) (i) Speed = $\sqrt{1.2^2 + 0.9^2 \sin^2 t}$

M1A1

2

(ii) Maximum speed = $\sqrt{1.2^2 + 0.9^2}$
Use of sint = 1

m1

= 1.5

A1

2

[6]

Q17.

(a) Using $F = ma$:
 $2400\mathbf{i} - 4800t\mathbf{j} = 800\mathbf{a}$

M1

$\mathbf{a} = 3\mathbf{i} - 6t\mathbf{j}$

A1

2

(b) $\mathbf{v} = \int \mathbf{a} \, dt$

M1

= $3t\mathbf{i} - 3t^2\mathbf{j} + \mathbf{c}$

Condone no '+ c'

A1

When $t = 0$, $\mathbf{v} = 6\mathbf{i} + 30\mathbf{j}$

$\therefore \mathbf{c} = 6\mathbf{i} + 30\mathbf{j}$

Needs '+ c' above

M1

$\therefore \mathbf{v} = (3t + 6)\mathbf{i} + (30 - 3t^2)\mathbf{j}$

AG

A1

4

(c) $\mathbf{r} = \int \mathbf{v} \, dt$

M1

= $\left(\frac{3}{2}t^2 + 6t\right)\mathbf{i} + (30t - t^3)\mathbf{j} + \mathbf{d}$

A1 i term, A1 j term; condone no '+ d'

A1,A1

When $t = 0$, $\mathbf{r} = 2\mathbf{i} + 5\mathbf{j}$

$$\therefore \mathbf{d} = 2\mathbf{i} + 5\mathbf{j}$$

M1

$$\therefore \mathbf{r} = \left(\frac{3}{2}t^2 + 6t + 2 \right)\mathbf{i} + (30t - t^3 + 5)\mathbf{j}$$

A1

5

[11]

Q18.

(a) $\mathbf{d} = 3\mathbf{i} - 6\mathbf{j}$

*Accept $\pm \mathbf{d}$ or displacement of 3, 6
shown on a diagram*

B1

$$3\mathbf{i} - 6\mathbf{j} = (\mathbf{i} - 2\mathbf{j})t$$

Or equivalent method for t

Accept ratio of vectors leading directly to ± 3

M1

$$t = 3$$

CAO

A1

3

(b) (i) $\mathbf{r} = (\mathbf{i} - 2\mathbf{j}) \times 4 + \frac{1}{2} \times 2\mathbf{j} \times 16$

M1

*full method for vector expression giving change
in position*

for correct subs (give $4\mathbf{i} + 8\mathbf{j}$)

A1

$$+6\mathbf{i} - 4\mathbf{j}$$

M1

$$= 10\mathbf{i} + 4\mathbf{j}$$

*ft slip provided obtain vector expression
($u = 0$ gives $6\mathbf{i} + 12\mathbf{j}$)*

A1F

4

(ii) $A(3,2) C(10,4)$

$$\mathbf{d} = 7\mathbf{i} + 2\mathbf{j}$$

*Attempt to find vector **AC** or **CA**
(using candidate's C)*

M1

$$|\mathbf{d}| = \sqrt{7^2 + 2^2}$$

$$AC = \sqrt{53} = 7.28$$

ft $\mathbf{d}$ provided two non-zero components

Accept $\sqrt{53}$

A1F

2

[9]

Q19.

(a) $\mathbf{v} = (6t^2 - 2t)\mathbf{i} + (1 - 12t^2)\mathbf{j}$

differentiating both components

M1

one component correct

A1

second component correct

A1

3

(b) (i) $\mathbf{v}\left(\frac{1}{3}\right) = \left(\frac{6}{9} - \frac{2}{3}\right)\mathbf{i} + \left(1 - \frac{12}{9}\right)\mathbf{j} = -\frac{1}{3}\mathbf{j}$

substituting the value for t into their $\mathbf{v}$

M1

correct velocity

A1

2

(ii) Travelling due south

correct description (Follow through from $\mathbf{v} = \pm k\mathbf{j}$)

A1ft

1

(c) $\mathbf{a} = (12t - 2)\mathbf{i} - 24t\mathbf{j}$

differentiating their velocity

M1

$\mathbf{a}(4) = 46\mathbf{i} - 96\mathbf{j}$

correct acceleration at time t

A1

correct acceleration at $t = 4$

A1

3

(d) $\mathbf{F} = 6(46\mathbf{i} - 96\mathbf{j}) = 276\mathbf{i} - 576\mathbf{j}$

apply Newton's second law correctly

M1

$F = \sqrt{276^2 + 576^2} = 639 \text{ N}$

finding magnitude

M1

correct magnitude

A1

3

or

$\alpha = \sqrt{46^2 + 96^2} = 106.45$

$F = 6 \times 106.45 = 639 \text{ N}$

[12]

Q20.

(a) $-\mathbf{i} + \mathbf{j} = 2\mathbf{i} - \mathbf{j} + 10\mathbf{a}$

Use of velocity equation

M1

$\mathbf{a} = -0.3\mathbf{i} + 0.2\mathbf{j}$

Correct equation

A1

Correct a

A1

3

$$(b) \quad \mathbf{r} = (2\mathbf{i} - \mathbf{j})t + \frac{1}{2}(-0.3\mathbf{i} + 0.2\mathbf{j})t^2 + 20\mathbf{i}$$

Use of constant acceleration equation for position

M1

$$= (2t - 0.15t^2 + 20)\mathbf{i} + (-t + 0.1t^2)\mathbf{j}$$

Correct i component

A1

Correct j component

A1ft

3

ft incorrect acceleration

$$(c) \quad (i) \quad \mathbf{r}(20) = (2 \times 20 - 0.15 \times 20^2 + 20)\mathbf{i} + (-20 + 0.1 \times 20^2)\mathbf{j} \\ = 0\mathbf{i} + 20\mathbf{j}$$

Substituting $t = 20$ into their expression for $\mathbf{r}$

M1

so due north of origin

Correct conclusion from correct working

A1

2

$$(c) \quad (ii) \quad \mathbf{v}(20) = 2\mathbf{i} - \mathbf{j} + 20(-0.3\mathbf{i} + 0.2\mathbf{j}) = -4\mathbf{i} + 3\mathbf{j}$$

$$\mathbf{v}(20) = \sqrt{4^2 + 3^2} = 5 \text{ ms}^{-1}$$

Finding velocity at $t = 20$

M1

Correct velocity

A1

Finding magnitude

m1

Correct speed

A1ft

4

ft incorrect acceleration

[12]

Q21.

- (a) Ball is a particle/no spin

B1: One assumption

B1

No air resistance/Only gravity or weight

B1: Second assumption

B1

2

- (b) (i) $24.5 t - 4.9t^2 = 0$

M1: Equation for vertical motion with height zero

M1

$$\left(t = 0 \text{ or } t = \frac{24.5}{4.9} = 5s \right)$$

AG

A1: Correct equation

A1

dM1: Solving for t

dM1

A1: Correct time from correct working

A1

4

- (b) (ii) $R = 10 \times 5 = 50 \text{ m}$

M1: Use of horizontal component of velocity to find the range

M1

A1: Correct range

A1

2

- (c) $20 = 10 t$

M1: Horizontal equation

M1

$$t = 2$$

A1: Time to reach wall

A1

$$h = 24.5 \times 2 - 4.9 \times 2^2 = 29.4 \text{ m}$$

*dM1: Vertical equation for height with
 $u = 24.5$ and a negative acceleration*

dM1

A1: Correct height

A1

4

- (d) No change as acceleration and initial velocity do not change with the mass

B1: No change

B1

B1: Explanation

B1

2

[14]

Q22.

Answers in (a), (b) and (c)(i) must be vectors

(a) $\mathbf{a} = 4t\mathbf{i} + 3\mathbf{j}$

Must be vector, both i and j present

B1

1

(b) $\mathbf{v} = 2t^2\mathbf{i} + 3t\mathbf{j}$

Integration of acceleration attempted

M1

ft $\mathbf{a}$, both i and j present

A1F

2

(c) (i) $\mathbf{r} = \int_0^t (2t^2\mathbf{i} + 3t\mathbf{j}) dt$

Integration attempted

M1

$$= \frac{2t^3}{3}\mathbf{i} + \frac{3t^2}{2}\mathbf{j}$$

*A1 each term. Condone i and j missing.
 -1 for constants present,
 -1 for both terms unsimplified*

A1A1

(ii) $\left. \begin{array}{l} \mathbf{v} \text{ parallel to } \mathbf{i} + \mathbf{j} \\ 2t^2 = 3t \end{array} \right\}$

M1

$$t = \frac{3}{2}$$

Solve, ft $\mathbf{v}$ with vector components

A1F

$$\mathbf{r} = \frac{9}{4}\mathbf{i} + \frac{27}{8}\mathbf{j}$$

A1F

$$\text{Distance} = \sqrt{\left(\frac{9}{4}\right)^2 + \left(\frac{27}{8}\right)^2}$$

Magnitude of $\mathbf{r}$ in any form

M1

$$= 4.06 \text{ (4.056)}$$

Numerical answer, ft candidate's $\mathbf{r}$ and t

A1F

5

[11]

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Q23.

(a) (i) $x = 14t$

Accept as components of position vector

Accept $x = 15.7 \cos 26.6$

B1

$$y = 7t - 4.9t^2$$

Use of equation

M1

3

Substitution

A1

Accept g ; accept $y = 15.7 \sin 26.6t - \frac{1}{2}gt^2$

(ii) $t = \frac{x}{14} \quad y = 7 \times \frac{x}{14} - 4.9 \times \left(\frac{x}{14}\right)^2$

Use of candidate's coordinates

M1

Substituted; accept unsimplified

A1

$$= \frac{x}{2} - 4.9 \frac{x^2}{196}$$

$$y = \frac{20x - x^2}{40}$$

AG; convincingly found

A1

3

(iii) $y = 0 \quad 20x - x^2 = 0$

Full method for range ($t = 1.4286$ for range,
or $t = 0.7143$ for greatest height and horizontal

distance then doubled, or $R = \frac{V^2 \sin 2\alpha}{g}$)

M1

$x = 20$

AWRT

A1

2

(b) $x = 12, y = 2.4$

Substitution into appropriate equation(s), ($t = 0.857$)

M1

For $y = 2.4$

A1

$\therefore$ under bar

ft provided $y > 0$

Alt: M1 for $x = 10$ subs and followed by comparison,

$H = 2.5 \text{ A1}$, conclusion A1

A1F

3

(c) Air resistance ignored, ball treated as particle...

B1

1

[12]

Q24.

(a) $\mathbf{v} = 6\mathbf{i} + 2t\mathbf{j}$

M1: differentiation attempted and vector quantity for $\mathbf{v}$ given

M1A1

2

(b) $sp = \sqrt{6^2 + 4t^2} \text{ ms}^{-1}$

M1: sum of squares attempted giving scalar expression

*A1 : all correct, accept $(2t)^2$
f.t. $\mathbf{v}$ with 2 components*

M1A1F

2

(c) $\sqrt{6^2 + 4t^2} = 6\sqrt{2}$

o.e.; scalar expression for $\mathbf{v}$ in terms of t from (b) used

M1

$$36 + 4t^2 = 36 \times 2$$

$$t = 3$$

f.t. minor slip in (b) provided t is positive solution of quadratic eqn

A1F

2

[6]

Q25.

(a) $\mathbf{r} = \int (4\mathbf{i} - 2t\mathbf{j}) \, dt$

attempted

M1

$$= 4t\mathbf{i} - t^2\mathbf{j} \quad (+ c)$$

A1

$$t = 0, \mathbf{r} = 8\mathbf{j}$$

used

m1

$$\mathbf{r} = 4t\mathbf{i} + (8\mathbf{i} - t^2)\mathbf{j}$$

A1F

4

(b) $t = 2, \mathbf{r} = 8\mathbf{i} + 4\mathbf{j}$

B1F

1

(c) $8 - t^2 = 0$

M1

$$t = 2\sqrt{2} \quad \text{or} \quad t = 2.83$$

A1

2

[7]

Q26.

(a) (i) vertical: $s = ut + \frac{1}{2}at^2$
 $0 = 7t - 4.9t^2$

Full method

M1

$$0 = t(7 - 4.9t)$$

$$0 = 7 - 4.9t$$

m1

$$t = \frac{10}{7} \text{ sec (1.43)}$$

A1

3

(ii) $OF = 21 \times \frac{10}{7}$

M1

$$= 30 \text{ m}$$

A1F

2

(b) vert : $0 = 3.5t - 4.9t^2$

Full method required

M1

$$t = \frac{3.5}{4.9}$$

m1

$$FG = 21 \times \frac{3.5}{4.9}$$

M1

$$= 15 \text{ m}$$

A1

4

(c) $GH = \frac{1}{2} \times 15 = 7.5$

B1F

$$OH = 30 + 15 + 7.5$$

M1

$$= 52.5 \text{ m}$$

A1F

3

[12]

Q27.

(a) $\Gamma = \int (2t\mathbf{i} + 4\mathbf{j}) dt$

Integration attempt for M1

M1

$$= t^2\mathbf{i} + 4t\mathbf{j} (+c)$$

A1

$$t = 0, \Gamma = 2j, 2j = c$$

m1

$$\Gamma = t^2i + (4t + 2)j$$

Any vector form

A1F

4

(b) $t = 2, \Gamma = 4i + 10j$

subs either value of t into r for M1

M1 A1F

$$t = 4, \Gamma = 16i + 18j$$

ft for vectors with 2 non-zero components

A1F

$$= 12i + 8j$$

subtraction of two vectors

A1F

4

Alternative:

$$[t^2i + 4tj]^2$$

M1

$$= (16i + 16j) - (4i + 8j)$$

A1F A1F

$$= 12i + 8j$$

A1F

subtraction

[8]

Q28.

(a) $x = 7t, y = 7t - \frac{1}{2} \times 9.8t^2 \quad \begin{pmatrix} 7t \\ 7t - 4.9t^2 \end{pmatrix}$

B1 horiz

M1 use of $s = ut + \frac{1}{2}at^2$

M1A1 for vert, accept 10 or g

B1M1A1

3

(b) $t = \frac{x}{7}$

B1

$$y = 7 \times \frac{x}{7} - \frac{1}{2} \times 9.8 \times \frac{x^2}{49}$$

M1A1

$$y = x - 0.1x^2$$

A1

4

Alternative (full verification method)

$x = 7t$ used B1

subs into $y = x - 0.1x^2$

M1A1

$y = 7t - 4.9t^2$ as in (a)

A1

Alternative: (use equation of path)

$$y = x \tan \alpha - \frac{gx^2}{2v^2 \cos^2 \alpha}$$

45° used

B1

subs

M1A1

simplify

A1

(c) $x = 4, \quad y = 4 - 0.1 \times 4^2$

or $t = \frac{4}{7} (0.571(43))$

M1

$y = 2.4$

A1

above wire as $2.4 > 2$

ft slip in y provided y positive

A1F

3

(d) $y = 0$
 $x - 0.1x^2 = 0$

M1

$$x(1 - 0.1x) = 0$$

m1 A1

$$x = 10$$

A1F

4

Alternative: (use of coordinate equations)

$$y = 0, 7t - 4.9t^2 = 0$$

M1

$$\Rightarrow t = \frac{10}{7} \quad (1.43)$$

m1A1

$$x = 7t = 10$$

A1F

(Accept 10.01 if $t = 1.43$ used)

Alternative: (use of symmetry)

$$v = 0, 0 = 7 - 9.8t$$

M1

$$t = \frac{5}{9.8}$$

A1

$$x = 7 \times t \times 2$$

m1

$$x = 10 \text{ (acc 9.996)}$$

A1F

Alternative: (use of equation for Range)

$$R = \frac{2V^2 \sin \alpha \cos \alpha}{g}$$

Correct formula used M1, 45° used m1

Correct subs A1,

result A1F, AWRT 10m, ft one slip.

(e) $\text{Speed}^2 = 7^2 + 7^2$

M1

$= 9.90 \text{ ms}^{-1}$

Accept 9.9, $7\sqrt{2}$,
or 9.91 if $t = 1.43$ used in downward motion

A1

2

[16]

Q29.

(a) $\mathbf{v} = (2t - 6)\mathbf{i} + t^2\mathbf{j}$

differentiation

M1 A1, A1

3

(b) $2t - 6 = 0$

M1

$t = 3$

ft differentiation for $t > 0$

A1F

2

(c) $\mathbf{a} = 2\mathbf{i} + 2t\mathbf{j}$

M1

$\mathbf{a}$ is function of t , not constant

Differentiation of 2 components of vector
expression for $\mathbf{v}$ attempted

CAO $\mathbf{a}$ correct and explanation correct;
reference to dependence on t required

A1

2

[7]

Q30.

(a) $-16\mathbf{i} + 16\mathbf{j} = \frac{1}{2} \mathbf{a} \times 4^2$

Use of vector constant acceleration equation

M1

Correct equation

A1

$$\mathbf{a} = \frac{-16\mathbf{i} + 16\mathbf{j}}{8} = -2\mathbf{i} + 2\mathbf{j}$$

Solving for a

M1

ag Correct a

A1

4

(b) $\mathbf{v} = 5(-2\mathbf{i} + 2\mathbf{j}) = -10\mathbf{i} + 10\mathbf{j}$

Expression for v at t = 5

M1

Correct expression

A1

$$v = \sqrt{10^2 + 10^2} = 14.1 \text{ ms}^{-1} \text{ (to 3 sf)}$$

Finding magnitude

M1

Correct speed

A1

4

(c) $20\mathbf{i} - 10\mathbf{j} + \mathbf{Q} = 3(-2\mathbf{i} + 2\mathbf{j})$

Application of Newton's second law to form equation

M1

Correct equation

A1

$$\mathbf{Q} = (-20 - 6)\mathbf{i} + (10 + 6)\mathbf{j} = -26\mathbf{i} + 16\mathbf{j}$$

Solving for Q

M1

Correct Q

A1

4

[12]

Q31.

(a) $\mathbf{v} = (3\mathbf{i} - 10\mathbf{j}) + (4\mathbf{i} + 2\mathbf{j})t$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

M1

$$= (3 + 4t) \mathbf{i} + (2t - 10) \mathbf{j}$$

ag Correct result from correct working

A1

2

(b) $2t - 10 = 0$

j component equal to zero

M1

$$t = 5$$

Correct time

A1

2

(c) (i) $\mathbf{r} = (3\mathbf{i} - 10\mathbf{j}) \times 10 + \frac{1}{2} (4\mathbf{i} + 2\mathbf{j}) \times 10^2$
Finding $\mathbf{r}$ when $t = 10$

Correct expression

M1

A1

$$= 230 \mathbf{i}$$

Correct final answer

A1

3

(c) (ii) $\mathbf{v} = (3 + 4 \times 10) \mathbf{i} + (2 \times 10 - 10) \mathbf{j}$
 $= 43\mathbf{i} + 10\mathbf{j}$

Correct velocity

B1

1

(d) $\mathbf{r} = 230\mathbf{i} + (43\mathbf{i} + 10\mathbf{j}) \times 10$

Uses zero acceleration

M1

Uses both answers from (c)

M1

$$= 660\mathbf{i} + 100\mathbf{j}$$

Correct expression for $\mathbf{r}$

A1

Correct simplified result

A1

$$r = \sqrt{660^2 + 100^2}$$

Finding magnitude

M1

$$= 668 \text{ m}$$

Correct distance

Follow through from part (c)

A1 ✓

6

[14]

Q32.

(a) $5\mathbf{i} - 2\mathbf{j} = 4\mathbf{i} + 3\mathbf{j} + 10\mathbf{a}$

Use of constant acceleration equation for velocity

M1

Correct equation

A1

$$\mathbf{a} = \frac{1}{10} (\mathbf{i} - 5\mathbf{j}) = (0.1\mathbf{i} - 0.5\mathbf{j}) \text{ ms}^{-2}$$

ag *Correct acceleration from correct working*

A1

3

(b) $\mathbf{r} = (4\mathbf{i} + 3\mathbf{j})t + 0.5(0.1\mathbf{i} - 0.5\mathbf{j})t^2$

Use of constant acceleration equation for position

M1

Correct expression

A1

2

(c) $\mathbf{r} = (4t + 0.05t^2)\mathbf{i} + (3t - 0.25t^2)\mathbf{j}$

j component is zero

M1

Correct equation

A1

$$3t - 0.25t^2 = 0$$

$$t(3 - 0.25t) = 0$$

Solving for t

m1

$$t = 0 \text{ or } t = \frac{3}{0.25} = 12 \text{ s}$$

$$t = 12 \text{ s}$$

Correct time

A1

4

(d) $\mathbf{v} = (4 + 0.1t)\mathbf{i} + (3 - 0.5t)\mathbf{j}$

$$3 - 0.5t = 0$$

$$t = 6$$

Expression for velocity

M1

Correct velocity

A1

j component is zero

m1

Correct time

A1

4

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[13]

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Q33.

(a) $60\mathbf{i} + 20\mathbf{j} = 20(2\mathbf{i} + 3\mathbf{j}) + 200\mathbf{a}$

$$20\mathbf{i} + 80\mathbf{j} = 200\mathbf{a}$$

$$\mathbf{a} = 0.1\mathbf{i} + 0.4\mathbf{j}$$

*Use of constant acceleration equation
in vector form to find a*

M1

Correct equation

A1

Correct a

A1

3

(b) $\mathbf{v} = (2\mathbf{i} - 3\mathbf{j}) + (0.1\mathbf{i} + 0.4\mathbf{j})t$
Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$= (2 + 0.1t)\mathbf{i} + (-3 + 0.4t)\mathbf{j}$
Correct expression

A1

2

(c) $2 + 0.1t = -(-3 + 0.4t)$
Equating components with $\pm$

M1

$0.5t = 1$
Correct equation

A1

$t = 2$

A1

$\mathbf{v} = 2.2\mathbf{i} - 2.2\mathbf{j}$
Finding velocity

M1

$v = \sqrt{2.2^2 + 2.2^2} = 3.11\text{ms}^{-1}$
Correct speed

A1

5

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[10]

Q34.

(a) $40\mathbf{i} - 15\mathbf{j} = \frac{1}{2} (5\mathbf{i} - 2\mathbf{j} + \mathbf{v}) \times 10$
Use of constant acceleration equation with $\mathbf{v}$ unknown
Correct equation

M1 A1

$\mathbf{v} = 8\mathbf{i} - 3\mathbf{j} - 5\mathbf{i} + 2\mathbf{j} = 3\mathbf{i} - \mathbf{j}$
Solving for $\mathbf{v}$
Correct $\mathbf{v}$

M1 A1

4

(b) $3\mathbf{i} - \mathbf{j} = 5\mathbf{i} - 2\mathbf{j} + 10\mathbf{a}$
Use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

M1A1

$\mathbf{a} = -0.2\mathbf{i} + 0.1\mathbf{j}$
Correct $\mathbf{a}$

A1

3

(c) $\mathbf{F} = 15(-0.2\mathbf{i} + 0.1\mathbf{j})$
Use of $\mathbf{F} = m\mathbf{a}$

M1

$= -3\mathbf{i} + 1.5\mathbf{j}$
Correct $\mathbf{F}$

A1

$F = \sqrt{3^2 + 1.5^2} = 3.35$
Finding magnitude of $\mathbf{F}$
Correct magnitude

M1 A1

4

[11]

Q35.

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(a) (i) $\mathbf{a} = \frac{1}{4}(8\mathbf{i} - 3\mathbf{j}) = 2\mathbf{i} - 3\mathbf{j}$
Use of $\mathbf{F} = m\mathbf{a}$, must be applied to both components

M1

Correct acceleration

A1

2

(ii) $\mathbf{v} = 20(2\mathbf{i} - 3\mathbf{j}) = 40\mathbf{i} - 60\mathbf{j}$
Use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$

M1

Correct $\mathbf{v}$

A1

2

(iii) $\mathbf{r} = \frac{1}{2} (2\mathbf{i} - 3\mathbf{j}) \times 20^2$

Use of constant acceleration equation to find $\mathbf{r}$ with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$

M1

Correct expression

A1

$= 400\mathbf{i} - 600\mathbf{j}$

Correct $\mathbf{r}$ in simplified form

A1

3

(b) $\mathbf{r} = 400\mathbf{i} - 600\mathbf{j} + 25(40\mathbf{i} - 60\mathbf{j})$

Use of $\mathbf{r} + 25\mathbf{v}$

M1

Correct expression

A1ft

$= 1400\mathbf{i} - 2100\mathbf{j}$

Correct position vector

A1ft

$r = \sqrt{1400^2 + 2100^2} = 2520 \text{ m (to 3 sf)}$

Finding magnitude

m1

Correct distance from correct working

A1ft

5

Alternative: Straight line method

M1 two distances r and $25v$

A1 for each distance

m1 adding

A1 correct final answer

[12]